

qicert: Certified Compression of Vision–Language–Action Models

with Exact Lipschitz Bounds and a Refusing Runtime Guard

Phase-1 Concept Proposal — 2026 Global Quantum + AI Challenge

Problem statement: *Quantum-Enhanced Vision-Language-Action Models in Autonomous Driving and Robotics Applications* (Volkswagen Group)

Code: <https://github.com/ZakLr/qicert> · Technical report: [docs/technical-report-v2.pdf](#) (7 pages, in repo)

1. Problem Framing

Vision–language–action models (VLAMs) for driving and robotics are 7B–10B parameter systems that must eventually run on embedded controllers under functional-safety regimes (ISO 26262, IEC 62061). The challenge statement names the compound constraint precisely: shrink the model *and* retain verifiable behavior on long-tail inputs. Classical compression solves the first half only. Quantization, pruning, and distillation reduce footprint, but none of them yields a statement of the form “under any input perturbation of size δ , the action cannot leave this set” — which is the artifact a safety engineer would need before certifying deployment on long-tail sensor distributions.

Quantum or quantum-inspired advantage enters where structure, not scale, does the work. Our QI component is tensor-network structure (tensor-train decomposition, the MPS/TTNS family the statement highlights). Its decisive property is not size reduction — at the ratios we tested it loses the accuracy contest to INT8 and we report that plainly — but that the compressed weights *carry an exact per-layer Lipschitz constant* computable from the compressed cores alone, with no reference to data. Quantized weights admit no comparable exact bound, because quantization error is data-dependent. Structure is therefore the classical-side object that makes an exact deployment certificate computable, chainable, and enforceable at runtime — and it is structurally unavailable to the accepted baseline. This is the specific, narrow sense in which the quantum-inspired method provides the credible advantage the guidelines ask for; we deliberately claim nothing broader.

2. Technical Approach

Paradigm: quantum-inspired (tensor-network), classical pipeline, no quantum hardware required or assumed. The statement’s simulation clause covers us explicitly; we state resources honestly in Section 3.

The pipeline wraps an existing vision–language–action model (VLA) at four points. A fine-tuning stage establishes the reference policy and freezes a hash-verified evaluation split. The compression stage decomposes each linear layer of the language backbone into a tensor-train network (TT-SVD); a closed-form low-rank residual folds the truncation loss back into the operator so that accuracy survives compression as well as the budget allows, with the certificate cost known exactly (triangle inequality on the operator norm). The certificate stage chains the exact per-layer bounds into a model-level perturbation ball and signs the artifact into a hash manifest. The deployment stage is a runtime guard that verifies the manifest at load time and, at every inference step, refuses any action outside the certified ball.

Three measured components complete the stack. First, a commuting-Pauli interaction compiler evaluates cross-modal interaction operators exactly over pruned families; its pruning predictions track measurement at median $r = 0.975$ over twelve independently generated operator test cases, with compiler identity verified to 2.3×10^{-14} relative error. Second, a rare-event estimation layer races three estimators on a known-ground-truth problem ($p^* = 1.37 \times 10^{-4}$): an iterated quantum amplitude estimation schedule (simulated; the estimator sees only measurement counts, as it would on hardware) reaches naive-Monte-Carlo interval width with $\sim 1,587\times$ fewer queries, and all arms cover the truth at budget. Third, a conformal layer calibrates violation thresholds whose empirical coverage lands inside the finite-sample band at all three tested levels ($\alpha = 0.01/0.05/0.10$).

Proof status. The layer bound $\|W\|_2 \leq \prod_k \|G^{(k)}\|_3$ is exact algebra on the tensor-train (TT) factorization; the certificate chain and the guard’s soundness/completeness properties are machine-checked by satisfiability-modulo-theories falsification (the Z3 solver finds no counterexample to accepting an out-of-ball action; key properties also proved in the Lean 4 proof assistant with no proof gaps). A guard demo transcript shows clean

load SERVE, out-of-ball action REFUSE, tampered-certificate REFUSE, NaN-bound REFUSE, missing-manifest REFUSE, plus an entropy gate flagging collapsed (single-token) streams.

Working prototype. This is not a design on paper. The repository contains 84 passing tests, a benchmark harness, and measured results on a fine-tuned 0.5B VLA (LIBERO-spatial, 114,497 scored action tokens, hash-frozen split): reference accuracy 0.4319 ± 0.0303 (seeds 0.4468, 0.3970, 0.4520); calibrated INT8 at $1.997\times$ compression, 0.4466 (the honest classical comparator); tensor-compressed routes at $2.5\text{--}2.9\times$ with sound certificates on every one of 168 compressed layers across every configuration tested. The pre-registered capability gate (ratio $\geq 2\times$, accuracy ≥ 0.3968 , fixed pre-run) is answered: the training-free route misses it (0.1640 at $2.536\times$ via single-token mode collapse), and repair training on the compressed model (arm N9, below) clears it (0.4496 ± 0.0137 at $2.46\times$, replicated across 3 seeds, certificates re-derived from the final deployed weights). Both verdicts, the negative one included, are reported in full in the technical report rather than hidden.

3. Feasibility and Resource Requirements

Feasibility is demonstrated, not projected: every component above has already run end to end on consumer hardware. The full experimental program to date consumed roughly 15 GPU-hours on a single laptop-class GPU (NVIDIA RTX 5060, 8 GB), including fine-tuning (1,200 steps), all compression arms, full-split evaluations, the estimator fixtures, and latency profiling. Peak training memory was 4.8 GiB. No proprietary data, cloud credits, or quantum hardware are involved at any stage; LIBERO is the statement’s own recommended robotics dataset; simulation environments and versions are pinned in the repository. Phase-2 scale-up (below) is sized for the same single-GPU discipline, with multi-seed replication the main added cost. The honest constraint we declare: tensor-network contraction on a CPU edge profile costs $3\text{--}15\times$ the dense fp16 kernel time per layer at these shapes — within the 100 ms budget on measured paths, but a real margin cost that Phase 2 must optimize rather than wish away.

4. Expected Impact

For the enterprise use case, the deliverable changes what deployment approval means. A fleet operator today ships a compressed model on the strength of benchmark scores; with qicert-style artifacts they would ship a signed certificate, a guard that refuses out-of-ball actions, and an audited statistical tail — the difference between “we tested it” and “it is bounded, and the machine enforces the bound.” The design tool built on our degradation measurements points the same direction: predict the largest safe compression *before* committing to full evaluation, turning post-hoc luck into engineering process.

Quantitative targets, anchored to what already runs: the reference VLA at 0.4319 ± 0.0303 and the INT8 comparator at $1.997\times$ define the bar the tensor route must clear; certificates already hold on every compressed layer at every tested ratio; the estimator layer already delivers the $\sim 1,587\times$ query reduction. The combined capability condition — at or beyond $2\times$ compression, within the pre-registered accuracy band, certificate-sound with guard enforcement — is already demonstrated at mean $\pm$ std over 3 seeds (N9: $2.46\times$, 0.4496 ± 0.0137 , GO). Phase 2 hardens it into an engineering result: larger backbones, closed-loop rollouts (mean $\pm$ standard deviation over 3 independent runs), activation-aware rank allocation to push the ratio higher at the same accuracy, backbone scale to widen the margin, and a collapse-boundary detector so the safe compression limit is predictable per model — the capability the quantized baseline cannot produce at any ratio.

5. Validation Plan

The Phase-2 sprint validates three claims, each with metrics fixed in advance (pre-registered in the repository before the runs they judge) — this is the strongest differentiator we can offer: the gates are committed, mechanical, and auditable.

Capability claim (Phase-1 verdict: GO via trained repair). The Phase-1 program ran the pre-registered gate (ratio $\geq 2\times$ AND accuracy ≥ 0.3968) to a two-part verdict. The training-free repaired tensor route is NO-GO: best certified point 0.1640 at $2.536\times$, with the failure mechanism identified (mode collapse to the modal action token; the agreement-vs-compression curve is bimodal, not gradual). The trained-repair route is GO: a rank-8 adapter trained 1,200 steps on train-split episodes only (frozen eval episodes excluded and verified), merged into the compressed model, re-evaluated on the full protocol — 0.4496 ± 0.0137 at

$2.46\times$ honest ratio over 3 seeds (0.4535, 0.4609, 0.4343; seed 0 alone: 95% Wilson interval [0.4506, 0.4564], non-overlapping above its reference’s [0.4439, 0.4497]), with certificates re-derived from the final deployed weights (168/168 exact) at a cost of one SVD per layer. Phase 2 escalates the remaining levers in cost order: activation-aware rank allocation, and backbone scale (a 1.5–7B backbone, whose weights carry the redundancy a 0.5B LoRA-tuned model lacks — at $2.5\times$ a 3B model retains more absolute capacity than the present model has in total). Metrics: pre-registered gate, already met at mean $\pm$ std over 3 seeds on both rows (baseline 0.4319 ± 0.0303 , repair 0.4496 ± 0.0137).

Safety claim. Metrics: certificate soundness (currently 168/168 across all configurations), guard refusal behavior (currently 1.0 on all four tampering scenarios plus in/out-of-ball cases), estimator coverage of known ground truth (currently 1.0 at budget for all arms), conformal coverage within finite-sample bands (currently true at all three levels).

Design-tool claim (Phase-1 negative result reported). The linear degradation predictor fitted to Phase-1’s measured agreement curve is vacuous ($R^2 = 0.16$) because the curve is bimodal — a mode-collapse signature, not a smooth degradation. Phase 2 replaces the linear model with a collapse-boundary detector (classify safe/collapsed rather than regress agreement) and revalidates on the scaled backbone; the Phase-1 curve and its mechanism are reported as the measurement that motivated this redesign.

Integration milestones for the sprint, in order: multi-seed replication; degradation-curve measurement and predictor validation; mixed-allocation search to the gate; edge-profile latency optimization for the tensor contraction (the known margin cost); and a hardware-pathway write-up estimating qubit count, circuit depth, and shot budget for the amplitude estimation component, which the challenge lists as a secondary objective and for which our simulation fixtures already define the interface.

6. Hybrid / Cross-Domain Integration

The design is hybrid by construction, and the division of labor is the point. Classical deep-learning infrastructure does the training and the heavy evaluation (PyTorch/CUDA on consumer GPU). The quantum-inspired tensor layer sits inside the model, where its algebraic structure is load-bearing: it is what makes the exact bound computable at all. Quantum algorithmic structure appears as the amplitude-estimation schedule for the statistical tail — executed in simulation against the same measurement-count interface it would have on hardware, so the estimator layer is hardware-swappable without redesign. Formal methods (SMT falsification via Z3, conformal calibration, scenario optimization) audit the whole stack mechanically rather than by argument. Cross-domain transfer is native to the method: the certificate machinery is backbone-agnostic and dataset-agnostic, so the same pipeline applies unchanged to the autonomous-driving track (the latency profile we measure is a CPU edge profile, and the challenge’s 100 ms real-time constraint is the same in both tracks); the robotics-track instantiation here was chosen because its accepted baseline is reproducible on our hardware.

7. Team Capability

The team combines the three disciplines this proposal straddles. The lead (Zaki) built and operates the full pipeline end to end: the fine-tuning harness, the tensor-compression kernel stack with its certificate algebra, the benchmark/ledger infrastructure with pre-registered gates, and the formal-methods integration — evidenced by the public repository’s measured results, 84-test suite, and reproducible artifact chain. Working practice: every number resolves to a run directory; eval splits frozen and hashed upfront; the training-free NO-GO reported first-class alongside the repair GO. AI agents leveraged under the lead’s direction; all claims traceable to committed artifacts.

Enclosures

- **Public repository:** <https://github.com/ZakLr/qicert> — README quickstart, pinned environment, 84-test suite, benchmark harness, experiment log, and every artifact behind the numbers above.
- **Technical report (8 pages):** [docs/technical-report-v2.pdf](#) in the repository — method, theory, measured results incl. the negative capability result, ablation isolating the QI component, limitations, and a claim-to-artifact provenance map (Appendix A) with seven artifact-generated figures.
- **Team profile:** submitted via the portal form (names, roles, affiliations, prior quantum/QI and domain experience) per guidelines §4.1.